

The spectral stress is an angle

A strengthening of Propositions 3 and 4

Insert for *The Companion-Matrix Projector Diagnostic*

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Abstract

The paper already computes $\Phi_\infty = 1 - (u_1 \cdot v_1)^2$ in five places without naming the quantity. It is $\sin^2$ of the principal angle between the transport's left and right dominant singular directions. Naming it costs one lemma and upgrades two propositions: the value of a polynomial continued fraction becomes the *tangent* of that angle, and the Kronecker composition law becomes the statement that *direction cosines multiply*. Figure 8's Φ scale becomes an angle scale on $[0^\circ, 90^\circ]$. Nothing in the paper changes; the same numbers acquire a geometric name. All claims verified to 60 digits; script PROP4-ANGLE-01.

1 The lemma

DROP-IN TEXT — new lemma, immediately before Proposition 3 (Section 9.1)

Lemma 1 (The spectral stress is an angle). *Let $\Phi(P) = \frac{1}{2} \| [P, P^\top] \|_F^2$ for a Frobenius-normalised transport product, and suppose P has collapsed to rank one with left and right dominant singular directions u_1, v_1 . Let $\theta \in [0, \pi/2]$ be the principal angle between $\text{span}(u_1)$ and $\text{span}(v_1)$. Then*

$$\Phi = \sin^2 \theta.$$

Proof. Write $P = u_1 v_1^\top$ with $\|P\|_F = 1$. Then $PP^\top = u_1 u_1^\top$ and $P^\top P = v_1 v_1^\top$, so $[P, P^\top] = u_1 u_1^\top - v_1 v_1^\top$ and $\|u_1 u_1^\top - v_1 v_1^\top\|_F^2 = 2 - 2(u_1 \cdot v_1)^2$. Hence $\Phi = 1 - (u_1 \cdot v_1)^2 = 1 - \cos^2 \theta$. $\square$

The identity $\Phi = 1 - (u_1 \cdot v_1)^2$ is the one already used throughout; the content of the lemma is only that the quantity so computed is a principal angle, and therefore an intrinsic Grassmannian invariant of the pair of one-dimensional subspaces the transport distinguishes. In particular it is unchanged under $P \mapsto cP$ for any $c > 0$, which is what the Frobenius normalisation in the definition of the diagnostic secures.

Verified for $d = 2, 3, 4, 6$: Φ against $\sin^2 \theta$ agrees to $0\text{--}8 \times 10^{-16}$.

2 Proposition 4 in angle form

DROP-IN TEXT — replacing the second sentence of Proposition 4 (Section 10.1)

Proposition 1 (The continued fraction is a tangent). *Let a polynomial continued fraction have a minimal solution and value K , so that $v_1(N) \rightarrow \text{normalize}(1, K)$. If moreover $a_n \rightarrow \infty$, so that $u_1 \rightarrow (1, 0)$, then in the notation of Lemma 1*

$$\tan \theta = |K|, \quad \text{equivalently} \quad \Phi_\infty = \frac{K^2}{1 + K^2} = \sin^2 \theta,$$

and the sign of K is carried by the orientation of v_1 .

So the diagnostic does not merely return the continued fraction: it returns it as the slope of one subspace against the other, and Φ_∞ is the squared sine of the angle between them. Confirmed on the three calibrated fractions whose hypothesis holds, to full working precision:

fraction	K	θ	$ K - \tan \theta $
Question 1 (π)	-0.375079835586	20.560055°	4×10^{-62}
e -fraction	0.581976706869	30.198408°	0
π -fraction	0.273239544735	15.282434°	0

DROP-IN TEXT — new remark, immediately after Proposition 1

Remark 1 (The hypothesis $a_n \rightarrow \infty$ is load-bearing). Apéry's $\zeta(3)$ recurrence has $a_n \rightarrow 34$, so $u_1 \rightarrow \text{normalize}(17 + 12\sqrt{2}, 1)$ rather than $(1, 0)$ and Proposition 1 does not apply. The measured value is $\Phi_\infty = 0.001442113945$, giving $\theta = 2.1763^\circ$, whereas $\arctan |K| = 0.4902^\circ$ for $K = 6/\zeta(3) - 5$. Lemma 1 still holds — Φ_∞ is still $\sin^2$ of the angle — but the angle is between v_1 and the true u_1 , not between v_1 and e_1 . Apéry's recurrence is the calibrated example that shows the hypothesis is not decorative.

3 Proposition 3 in angle form

DROP-IN TEXT — new corollary to Proposition 3 (Section 9.1)

Proposition 2 (Direction cosines multiply). *For transport sequences $A(n), B(n)$ to which the diagnostic applies and $T(n) = A(n) \otimes B(n)$, the composition law $1 - \Phi_\infty^{A \otimes B} = (1 - \Phi_\infty^A)(1 - \Phi_\infty^B)$ is precisely*

$$\cos \theta_{A \otimes B} = \cos \theta_A \cos \theta_B.$$

Equivalently, in terms of the continued-fraction values, $K_{A \otimes B}^2 = K_A^2 + K_B^2 + K_A^2 K_B^2$.

This is the classical rule for angles between subspaces of a tensor product, and it says the non-normality of a Kronecker product is the geometric composition of its factors' angles rather than an unrelated bookkeeping law. Verified on Question 3: $\theta_e = 30.198408^\circ$, $\theta_\pi = 15.282434^\circ$, $\theta_{Q_3} = 33.516584^\circ$; the product of the first two cosines matches the third to 8×10^{-62} , and $\tan \theta_{Q_3} = 0.6623018907912486$ agrees with $\sqrt{K_e^2 + K_\pi^2 + K_e^2 K_\pi^2}$ to the same precision.

4 Figure 8 becomes an angle scale

DROP-IN TEXT — replacement caption / accompanying sentence for Figure 8

By Lemma 1 the spectral-stress scale is an angle scale. Each calibrated transport occupies a position in $[0^\circ, 90^\circ]$ according to how far its left and right dominant directions have separated: Question 8's transport becomes asymptotically normal, and its angle vanishes; Problem 2.5's reaches exactly 90° , at which the two directions are orthogonal.

system		Φ_∞	θ	
Q8	$\sqrt{10005}/\pi$	0	0.000°	asymptotically normal
Apéry	$\zeta(3)$	0.0014421139	2.176°	
Q1	π (PCF)	0.1233336964	20.560°	
Q3	$\pi + e$	0.3049009047	33.517°	Kronecker
2.6	$\zeta(2) + \zeta(3)$	2809/4178	55.081°	exact rational Φ_∞
2.7	$\zeta(2) + \zeta(3)$	0.7618951393	60.793°	tied recessive pair
Q2	γ	0.9442325877	76.341°	
2.5	Catalan G	1	90.000°	$u_1 \perp v_1$

Adding Apéry’s recurrence to the figure is free and worth doing: it is the one calibrated transport that is almost normal without being trivially so, and it sits between Question 8 and Question 1 where the figure is currently empty.

5 End-to-end check on Question 1

Running the real transport and measuring the angle directly against the closed form built from the official proof’s exact anchors:

N	measured θ	$ \theta - \arctan K $
50	20.5600534304°	3.5×10^{-8}
200	20.5600553769°	5.6×10^{-10}
800	20.5600554083°	8.8×10^{-12}
closed form	20.5600554088°	$K = 6(22 - 7\pi)/(3 - \pi)$

6 Where this goes, and what it does not change

- Lemma 1: new, immediately before Proposition 3 in Section 9.1.
- Proposition 1: replaces the second sentence of Proposition 4, Section 10.1.
- The $a_n \rightarrow \infty$ remark: immediately after it. This is a strengthening, not a correction — the existing statement already carries the hypothesis.
- Proposition 2: a corollary to Proposition 3, Section 9.1.
- Figure 8: caption sentence, plus the Apéry row.
- Summary item 7 gains the tangent statement; item 6 gains the cosine law.
- Appendix: PROP4-ANGLE-01 as a fifteenth listing.

No numerical value in the paper changes. No proof is corrected. Every quantity in the tables above is one the paper already reports; what is new is the name of the quantity and the two composition laws that follow once it has one.

One structural remark worth making in the Summary: the same invariant appears in three otherwise unrelated sectors of this work — the principal angle here, the $\cos^2$ principal angles that index a boundary channel census, and the $\sin^2$ form of the refraction law. The Hilbert–Schmidt distance between two projectors is $2 \sum_i \sin^2 \theta_i$ by construction, so any diagnostic built from projector distances is natively $\sin^2$ -valued in principal angles. That is not a coincidence between the sectors; it is the same measurement performed on different pairs of subspaces.